Learning Method for Extraction of Partial Correspondence from Parallel Corpus

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***Abstract*—For machine translations using a parallel corpus, it is effective to extract partial correspondences: pairs of phrases of the source language(SL) and target language(TL) in bilingual sentences. However, it is difficult to extract the partial correspondences correctly and efficiently in the data sparse corpus. In this paper, we propose a new learning method that extracts the partial correspondences solely from the parallel corpus without any analytical tools. In the proposed method, the extraction rules are automatically acquired from bilingual sentences using bi-gram statistics in each language sentence and the similarity based on Dice coefficient between SL words and TL words. The acquired extraction rules possess information about the first parts(*e.g.*, "a", "the") or the last parts in phrases. Moreover, the partial correspondences are extracted from the bilingual sentences using the extraction rules correctly and efficiently. Evaluation experiments indicated that our proposed method can improve the translation quality of the learning-type machine translation by correctly and efficiently extracting the partial correspondences in bilingual sentences.**

***Keywords*-extraction rule; learning; machine translation; partial correspondence; parallel corpus;**

I. Introduction

Recently, the machine translations using a parallel corpus have been researched actively. In those researches, it has been effective to extract the partial correspondences, which are pairs of the phrase in source language(SL) and the phrase in target language(TL) in bilingual sentences. Two approaches can be used to extract the partial correspondences: an analytical approach[1] or a statistical approach[2][3][4][5]. In the analytical approach, the systems depend on static language knowledge(*e.g.*, bilingual dictionary, POS, syntax analyzer). Therefore, it is difficult to extract the partial correspondences in languages for which analytical tools do not exist. In the statistical approach, the systems require a large parallel corpus to obtain the correct partial correspondences in bilingual sentences. That is, the systems based on the statistical approach have the sparse data problem.

As described in this paper, we propose a new learning method that efficiently and correctly extracts the partial correspondences, which are pairs of SL phrases and TL phrases, only from the parallel corpus without any analytical tools. In the proposed method, the extraction rules are first acquired automatically from bilingual sentences. The extraction rules are acquired using bi-gram statistics for words in each language sentence and the similarity based on Dice coefficient between SL words and TL words. Moreover, the extraction rules possess information about the first parts(*e.g.*, "a", "the" in English) or the last parts in the phrases. Therefore, the system with the proposed method can extract the partial correspondences in the bilingual sentences both correctly and efficiently. In order to confirm the effectiveness of the proposed method, we applied it to machine translation system based on learning method[6], which acquires translation rules solely from a parallel corpus automatically and performs translation using the acquired translation rules. The reason for it is that we can confirm effectiveness of our proposed method from the perspective of learning ability by using such a learning-type machine translation system. Evaluation experiments showed that the scores of some automatic evaluation for the system with the proposed method were 0.026–0.047 higher than those of the system without the proposed method. Moreover, in noun phrases that occur with low frequency, the extraction rate of them in the system with the proposed method was respectively 18.44 and 12.14 points higher than those in the system without the proposed method and the phrase-based SMT system. Therefore, we confirmed that our proposed method is effective to improve the translation quality of the learning-type machine translation by correctly and efficiently extracting the partial correspondences in bilingual sentences.

II. Outline of Learning-type Machine Translation System

We apply our proposed method to the learning-type machine translation system[6]. Figure 1 shows the outline of the learning-type machine translation system. In the translation process, the system generates translation results using translation rules acquired in the learning process. The user then proofreads the translated sentences, checking for errors. In the feedback process, the system evaluates the translation rules used in the translation process. In the learn-

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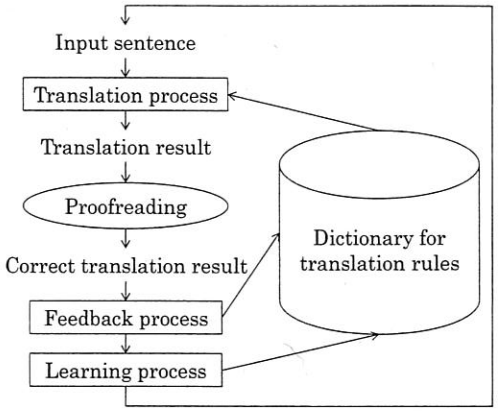


Figure 1. Outline of processing.

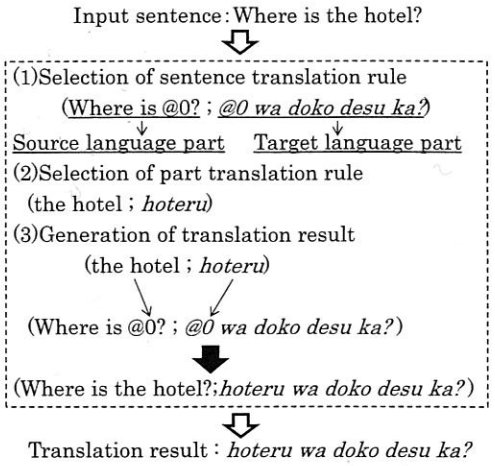


Figure 2. Example of generation of the translation result.

ing process, the translation rules are acquired automatically. In Fig. 1, dictionary for translation rules has translation rules acquired only by the learning process, without using any static language knowledge(*e.g.*, bilingual dictionary).

A. Translation Process

In the translation process, the system generates translation results for input sentences. Figure 2 shows an example of generation of a translation result. First, the system selects the sentence translation rules that can be applied for input sentence. Second, the system selects the part translation rules. Herein, the sentence translation rules have the bilingual structure of a sentence such as (Where is @0?;*@0 wa doko desu ka*[1]*?*). The part translation rules mean the bilingual parts such as (the hotel;*hoteru*). The system generates the translation results by substitution the variables in the sentence translation rules with the part translation rules. As a result, "*hoteru wa doko desu ka?*" is obtained as the translation result.

B. Feedback Process

In the feedback process, the system evaluates the translation rules used in translation process. When the translation rules are used to generate the correct translation results, the system increases the correct translation frequencies of the used translation rules. When the translation rules are used to generate the erroneous translation results, the system increases the erroneous translation frequencies of the used translation rules.

C. Learning Process

In the learning process, the system acquires the translation rules using Inductive Learning.

[1]Italics expresses the pronunciation in Japanese.

Bilingual sentence 1
(I'd like a room with a balcony. ;
barukonî tsuki no heya wo onegai shi masu.)
Bilingual sentence 2
(Can I get a room for tonight?;
kon'ya heya wa ari masu ka?)
⇩ Extraction of a common part
Part translation rule : (a room;*heya*)
⇩ Substitution of variables "@0" for common parts
Sentence translation rules
(I'd like @0 with a balcony.;
barukonî tsuki no @0 wo onegai shi masu.)
(Can I get @0 for tonight?; *kon'ya @0 wa ari masu ka?*)

Figure 3. Example of acquisition of translation rules using Inductive Learning.

1) Inductive Learning: In Inductive Learning, the translation rules are acquired by extracting a common part and a different part in the bilingual sentence pairs. Figure 3 shows an example of acquisition of translation rules using Inductive Learning. In bilingual sentences 1 and 2, (a room;*heya*), which is a common part, is extracted and it becomes the part translation rule. Moreover, the system acquires (I'd like @0 with a balcony.;*barukonî tsuki no @0 wo onegai shi masu.*) and (Can I get @0 for tonight?;*kon'ya @0 wa ari masu ka?*) as the sentence translation rules by replacing the common part (a room;*heya*) in the bilingual sentences 1 and 2 with variables "@0". Moreover, the system acquires more general translation rules applying Inductive Learning to the acquired pairs of translation rules.

III. Learning Method for Extraction of Partial Correspondence

In the proposed method, the system acquires the extraction rules using bi-gram statistics for words of each language

sentence and the similarity based on Dice coefficient between SL words and TL words. The extraction rules possess information to extract the partial correspondences in the bilingual sentences. For example, using the extraction rule (a @0 room;*@0 heya*), the system extracts the part from "a" to "room" in the SL sentences because the variable "@0" exists between "a" and "room" in this extraction rule. Moreover, the system extracts the part which consists of "*heya*" and the part which adjoins the left side of "*heya*" from TL sentences because the "@0" exists the left side of "*heya*" in this extraction rule. This means that this extraction rule possesses the information for extracting the partial correspondence. Therefore, the system can extract the phrase in a sentence both efficiently and correctly. The extraction rules are two kinds: one is the first extraction rules which is acquired using bi-gram statistics and the similarity, and the other is the second extraction rules which is acquired using the first extraction rules.

A. Acquisition of the First Extraction Rules

It is necessary for the system to extract the phrase-level correspondence, not only the word-level correspondence. Therefore, the system acquires (a @0 room;*@0 heya*) as the first extraction rule. In this first extraction rule, the parts between "a" and "room" mean the parts that can be delete in English sentences. As the result, the system can determine the phrase-level correspondence in bilingual sentences. Moreover, all extraction rules are acquired without using any analytical tools.

Figure 4 shows an example of acquisition of the first extraction rule. The system selects a bilingual sentence which includes both the SL part and TL part in the part translation rule acquired by Inductive Learning. In Fig. 4, the system selects the bilingual sentence (I'd like a quiet room with a bath.;*basu tsuki no shizuka na heya wo onegai shi masu.*) when (a room;*heya*) is acquired already as the part translation rule because both the SL part "a room" and TL part "*heya*" in the part translation rule (a room;*heya*) are included in the bilingual sentence (I'd like a quiet room with a bath.;*basu tsuki no shizuka na heya wo onegai shi masu.*).

The system extracts the phrase parts from each language sentence. When the number of the common parts is 2, the system extracts the part between two common parts and two common parts. In Fig. 4, "quiet" between "a" and "room" in the SL sentence is determined as the deletion-possible part by the SL part "a room" in the part translation rule, and "a quiet room" is extracted as the phrase part. The deletion-possible part is determined using bi-gram statistics for words when the number of the common parts is only 1. In Fig. 4, the common part is only "*heya*" in TL sentence. Therefore, the system determines the deletion-possible part by calculating bi-gram statistics between "*heya*" and another word using all given bilingual sentences. The system calculates bi-gram statistics between "*heya*" and the word "*shizuka*"

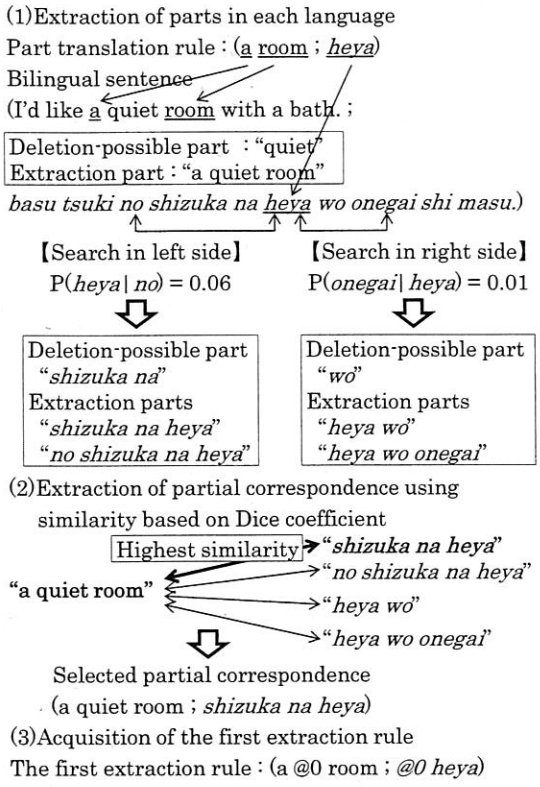


Figure 4. Example of acquisition of the first extraction rule.

of the left side of "*heya*". As the result, bi-gram statistics between "*heya*" and "*shizuka*" is 0.00. The system continues the calculation of bi-gram statistics, until bi-gram statistics becomes over 0.00 using "*heya*" and other words. The system calculates bi-gram statistics between "*heya*" and the word "*no*" of the next left side of "*heya*". In the case, the bi-gram statistics between "*heya*" and "*no*" is 0.06. Therefore, the deletion-possible part is "*shizuka na*" between "*heya*" and "*no*". Moreover, the system extracts the parts using the common part "*heya*", the deletion-possible part "*shizuka na*" and the word "*no*", which was used to calculate bi-gram statistics. That is, the system extracts "*shizuka na heya*" and "*no shizuka na heya*" from TL sentence. In the right side of the common part "*heya*", the system extracts the parts "*heya wo*", "*heya wo onegai*" using bi-gram statistics.

Next, the system determines the partial correspondence using "a quiet room" extracted from SL sentence and "*shizuka na heya*", "*no shizuka na heya*", "*heya wo*" and "*heya wo onegai*" extracted from TL sentence. In that case, the system uses the similarity based on Dice coefficient between SL words and TL words. As the result, the system determines (a quiet room;*shizuka na heya*), which has highest similarity, as the partial correspondence. Moreover, the first

extraction rule is acquired by replacing the deletion-possible parts with variables "@0" in the partial correspondence. In Fig. 4, the system replaces the deletion-possible part "quiet" in the extracted "a quiet room" with variable "@0" in SL, and replaces the deletion-possible part *"shizuka na"* in the extracted "*shizuka na heya*" with the variable "*@0*" in TL. Consequently, (a @0 room;*@0 heya*), that is the pair of "a @0 room" and "*@0 heya*", is acquired as the first extraction rule.

Moreover, the system acquires (a @0;*@0*) as the more general first extraction rule by extracting the different parts between two first extraction rules (a @0 room;*@0 heya*) and (a @0 hotel;*@0 hoteru*) when (a @0 hotel;*@0 hoteru*) has already been acquired from other bilingual sentences.

B. Acquisition of Translation Rules using the First Extraction Rules

The system acquires the part and sentence translation rules using the first extraction rules. Figure 5 shows an example of acquisition of the translation rules using the first extraction rule. First, the system selects the bilingual sentences which includes both SL parts and TL parts of the first extraction rules. In Fig. 5, the system selects (a @0;*@0*) as the first extraction rule because "a", which is the SL part of the first extraction rule, is included the bilingual sentence (Do you have a return ticket?;*kaeri no koukuu ken wo o moti desu ka?*).

Next, the system determines the partial correspondence in the selected bilingual sentence using the common parts. In Fig. 5, "return", which adjoins the right side of "a", is selected from SL sentence because variable "@0" exists right side of "a" in SL part of the first extraction rule. The system selects "*kaeri*" which corresponds to "return" from TL sentence because the similarity based on Dice coefficient between "return" and "*kaeri*" is over threshold based on preliminary experiment(*i.e.*,0.35) and the highest similarity among the similarities between "return" and other words in TL sentence. Moreover, "ticket", which adjoins the right side of "return", is selected. The system selects "*ken*" which corresponds to "ticket" from TL sentence because the similarity between "ticket" and "*ken*" is over threshold and the highest similarity among the similarities between "ticket" and other words in TL sentence. Therefore, the system extracts "*kaeri no koukuu ken*", which is the part from "*kaeri*" to "*ken*", from TL sentence. As the result, the system extracts (a return ticket;*kaeri no koukuu ken*) as the partial correspondence combining "a return ticket", which consist of the common part "a" and "return ticket" extracted from SL sentence, and "*kaeri no koukuu ken*" extracted from TL sentence.

Finally, the system acquires the translation rules using the partial correspondences. In Fig. 5, (a return ticket;*kaeri no koukuu ken*), which is the partial correspondence in the bilingual sentence, is acquired as the part translation rule.

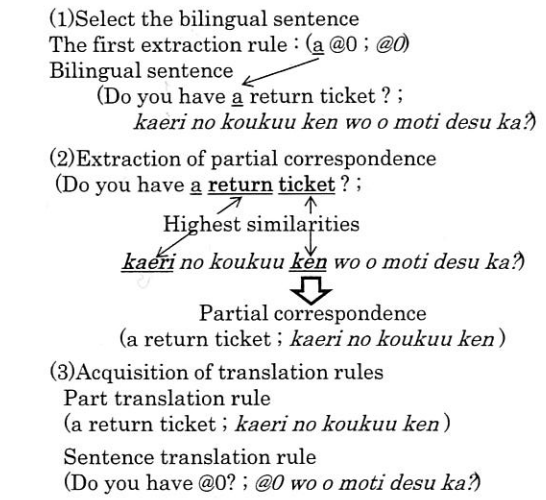


Figure 5. Example of acquisition of translation rules using the first extraction rule.

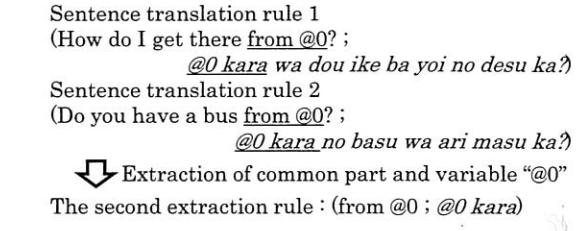


Figure 6. Example of acquisition of the second extraction rule.

Moreover, (Do you have @0? ; *@0 wo o moti desu ka?*) is acquired as the sentence translation rule replacing the partial correspondence (a return ticket;*kaeri no koukuu ken*) with variables "@0".

In the proposed method, the partial correspondence in the bilingual sentences can be extract using the extraction rules correctly because the extraction rules have the information about the first parts(*e.g.*, "a") or the last parts in partial correspondence.

C. Acquisition and Application of the Second Extraction Rules

In the proposed method, the system acquires the second extraction rules from the sentence translation rules acquired using the first extraction rules. Figure 6 shows an example of acquisition of the second extraction rule. In the sentence translation rules 1 and 2, the system acquires (from @0;*@0 kara*) as the second extraction rule combining "from @0", which consists of the common part "from" and the variable "@0" between two SL parts, and "*@0 kara*", which consists of the common part "*kara*" and the variable "*@0*" between two TL parts. This second extraction rule (from @0;*@0*

Table I
EXPERIMENTAL RESULTS OBTAINED USING SYSTEMS WITH AND WITHOUT THE PROPOSED METHOD.

System	BLEU	IMPACT	ROUGE-L	mWER	mPER
With Proposed Method	0.355	0.549	0.560	0.555	0.460
Without Proposed Method	0.329	0.519	0.525	0.585	0.507

Table II
EFFECTIVENESS OF PROPOSED METHOD IN TRANSLATION RESULTS.

	Input sentence	Without Proposed Method	With Proposed Method
No.1	I'd like a room with shower.	*tsuki no shawâ heya wo onegai si masu.*	*shawâ tsuki no heya wo onegai si masu.*
No.2	Can I have a room with a shower?	a shower *beddo tsuki no heya wa ari masu ka?*	*shawâ tsuki no heya wa ari masu ka?*
No.3	How much will it take to Sheraton Hotel?	How much will it take to Sheraton Hotel? (Translation impossibility)	*sheraton hoteru made ryoukin wa dono kurai kakari masu ka?*

kara) means that the parts which adjoins the right side of "from" in SL sentences correspond to the parts which adjoins the left side of "*kara*" in TL sentences.

The system extracts the partial correspondence using the second extraction rules. In this case, only parts which correspond to the variables "@0" are used as the partial correspondence. That is, the common parts are not included in the partial correspondence. For example, when the second extraction rule (with @0;*@0 tsuki no*) is applied to bilingual sentence (Can I have a room with twin beds?;*tsuin beddo tsuki no heya wa ari masu ka?*), (twin beds;*tsuin beddo*), which corresponds to the variables "@0", is extracted as the partial correspondence. Moreover, the system acquires (twin beds;*tsuin beddo*) as the part translation rule and (Can I have a room with @0?;*@0 tsuki no heya wa ari masu ka?*) as the sentence translation rule replacing the extracted partial correspondence with the variables "@0".

IV. EXPERIMENTS

A. Experimental Procedure

In this paper we performed the evaluation experiments to confirm the effectiveness of the proposed method. In that case, we used two MT systems. One is the system with proposed method described in sections II and III, and the other is the system without proposed method described in II. These systems translated English sentences into Japanese sentences using an experimental data. There are two kinds of data as the experimental data: learning data and evaluation data. In these experiments, 948 bilingual sentences were used as learning data. As well, 762 bilingual sentences were used as evaluation data. These bilingual sentences were taken from an English–Japanese travel conversation text. All of these bilingual sentences were processed using the method outlined in Fig. 1. The initial condition of the dictionary is empty.

B. Evaluation Method

We used BLEU, IMPACT[8], ROUGE-L, mWER, and mPER as the automatic evaluation to evaluate two systems. In this case, four Japanese sentences by bilingual humans were used for each translated Japanese sentence to calculate the scores. In mWER and mPER, the evaluation for MT systems is high when the score is low.

Table III
EXAMPLE OF EFFECTIVE EXTRACTION RULES.

The first extraction rules	The second extraction rules
(a @0 room;*@0 heya*)	(with @0;*@0 tsuki no*)
(the @0 Hotel;*@0 hoteru*)	(near @0;*@0 no chikaku*)
(a @0;*@0*)	(at @0;*@0 ni*)
(the *@0;@0*)	(from @0;*@0 kara*)

C. Experimental Results

Table I shows the scores of the systems with and without the proposed method respectively. In Table I, the scores of the system with proposed method improved 0.026–0.047 to the system without the proposed method.

Moreover, Table II shows examples of the translation results that improve from the erroneous translation results to the correct translation results by the proposed method. In example No.1 of Table II, the error of the word order is included in the translation result of the system without proposed method. In example No.2 of Table II, "a shower" is not able to be translated, and "*beddo*", means "bed" in English, is included in the translation result of the system without proposed method. In example No.3 of Table II, translation impossibility means that the MT system was not able to translate the input sentence at all because the sentence translation rule, that can be applied to the input sentence, did not exist at all. In the system with the proposed method, these input sentences were translated into the correct translation results. The reason for it is that the proposed method improved the learning ability of the system. As the result, the system with the proposed method could acquire more effective translation rules.

D. Discussion

In Table I, we confirmed that the scores of the system with the proposed method is higher than those of the system without the proposed method. The reason for it is that the effective translation rules increased using the proposed

method. In the experiments, the effective sentence translation rules increased by 885, and the effective part translation rules increased by 733. In this paper, the effective translation rules mean that the SL parts correspond to the TL parts correctly.

Next, we describe the extraction rules acquired by the proposed method. Table III shows the example of the effective extraction rules. By this experiment, 25 first extraction rules and 58 second extraction rules were acquired in the system with proposed method. Of them, the numbers of the effective first extraction rules and the effective second extraction rules are 13 and 16 respectively.

In this proposed method, the erroneous extraction rules are also acquired because the system uses the erroneous part translation rules acquired by Inductive Learning. For example, in the erroneous part translation rule (into dollars;*doru*) and the bilingual sentence (Could you change 100 Canadian dollars into U.S. dollars, please?;*100 kanada doru wo US doru ni ryougae shi te kudasai?*), "U.S." in the SL sentence and "*US*" in the TL sentence were determined as the deletion-possible parts. As the result, (into @0 dollars;*@0 doru*) was acquired as the erroneous first extraction rule. In this extraction rule, "into" in SL part must be deleted. To solve this problem, we need to improve the feedback process in order to evaluate the translation rules sufficiently.

Moreover, we investigated the extraction rate of noun phrases for which the frequency is only 1 in all experimental data(1,710 bilingual sentences) from the perspective of the data sparse problem. As the results, the extraction rates of the system with the proposed method and the system without the proposed method are $30.88\%(=\frac{201}{651}\times 100)$ and $12.44\%(=\frac{81}{651}\times 100)$, respectively. On the other hand, the extraction rate of the phrase-based statistical machine translation(SMT) system is $18.74\%(=\frac{122}{651}\times 100)$. The phrase-based SMT system was built using GIZA++[2], SRILM[3], and Moses[4], and it used the experimental data same as the experimental data used in our MT systems. These results show that the proposed method is effective to the data sparse problem.

V. Related Work

Ong[9], Malavazos[10] and Güvenir[11] proposed the method for acquiring the translation rules by extracting the common part and different part in the bilingual sentence pair. However, in these methods, the similar bilingual sentence pairs are required in order to extract the correct partial correspondences. That is, these method have the data sparse problem. Moreover, Chiang[5] proposed the method which acquires the translation rules by extracting the partial correspondences using GIZA++. However, it requires a large parallel corpus to the correct partial correspondences. For these problems, using the extraction rules acquired only from bilingual sentences without any analytical tool, our proposed method extracts the partial correspondences in the bilingual sentences both correctly and efficiently.

VI. Conclusion

In this paper, we proposed a new learning method which extracts the partial correspondences in bilingual sentences correctly and efficiently without any analytical tools. By the proposed method, the scores of automatic evaluation in the learning-type machine translation system improved 0.026–0.047. The extraction rate of low-frequency noun phrases in the system was also improved 12.14–18.44 points. These results indicate that the proposed method is effective at alleviating the sparse data problem.

In the future, we plan to apply our proposed method for the phrase-based statistical machine translation.

References

[1] M. Sasayama, F. Ren, and S. Kuroiwa, Automatic Super-Function Extraction for Translation of Spoken Dialogue, Natural Language Processing and Knowledge Engineering 2007, pp.141-148, 2007.

[2] F.J. Och and H. Ney, A Systematic Comparison of Various Statistical Alignment Models, Computational Linguistics, Vol.16, No.2, pp.79-85, 2003.

[3] A. Stolcke, SRILM – an Extensible Language Modeling Toolkit, 7th International Conference on Spoken Language Processing, pp.901-904, 2002.

[4] P. Koehn, H. Hoang, A. Birch, C. Callison-Burch, M. Federico, N. Bertoldi, B. Cowan, W. Shen, C. Moran, R. Zens, C. Dyer, O. Bo jar, A. Constantin, and E. Herbst, Moses: Open Source Toolkit for Statistical Machine Translation, Proc. ACL 2007, pp.177-180, 2007.

[5] D. Chiang, Hierarchical Phrase-Based Translation, Computational Linguistics, Vol.33, No.2, pp.201-228, 2007.

[6] H. Echizen-ya, K. Araki, Y. Momouchi and K. Tochinai, Study of Practical Effectiveness for Machine Translation Using Recursive Chain-link-type Learning, Proc. COLING 2002, pp.246-252, 2002.

[7] K. Araki and K. Tochinai, Effectiveness of Natural Language Processing Method Using Inductive Learning, Proc. the IASTED International Conference ARTIFICIAL INTELLIGENCE AND SOFT COMPUTING, pp.295-300, 2001.

[8] H. Echizen-ya, K. Araki, Automatic Evaluation of Machine Translation based on Recursive Acquisition of an Intuitive Common Parts Continuum, Proc. 11th Machine Translation Summit, pp.151-158, 2007.

[9] E. Ong, K. Go, M. Morga, V. Nunez, and F. Veto, Extracting and Using Translation Templates in an Example-Based Machine Translation System, Journal of Research in Science, Computing, and Engineering, pp.83-100, 2007.

[10] C. Malavazos and S. Piperidis, Application of analogical modeling to example based machine translation, Proc. Coling 2000, pp.516-522, 2000.

[11] H.A. Güvenir, and I. Cicekli, Learning translation templates from examples, Information Systems, Vol.23, No.6, pp.353-363, 1998.